

Development of Autonomous Wireless Beehive Monitoring System Solution

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Abstract—The autonomous beekeeping solution provides the possibility to monitor the bee hives from a distance. The distance monitoring opens a floor for new business opportunities, like production of mono-floral honey, as well as more intensive manufacture of honey products. The solution is based on new generation of IoT resources, and deployment of IoT concept in agricultural sector [3; 23; 24; 25]. It is worth noting that physical survey disturbs the bees and causes them stress, which can negatively affect the productivity of the bee colony, as well as taking beekeeper time. In addition, distance does not always allow the beekeeper to react quickly to crisis situations in the apiary. The technology developed in the project allows to use distance monitoring of the beehive, which significantly facilitates the daily life of the beekeeper, allows quick decision-making in critical situations and does not reduce bee productivity, as it does not affect the daily life of the beehive.

The beekeeper can observe the apiary remotely and will be able to decide to visit the apiary only if necessary. A great helper is the hive video surveillance system, which will help to monitor the flight activity of bee colonies during the day, as well as to visually assess the condition of the apiary. The new system uses alternative energy sources (solar, wind energy) for the power supply of the system, which increases the possibilities of using the system in distance monitoring on fields.

Keywords—autonomous beekeeping, beehive monitoring, monitoring, protection, system solution, video

I. INTRODUCTION

The creation of a reliable autonomous monitoring system to produce bee products in different conditions is especially important for increasing productivity of bee farms, as well as for identifying emergency situations in the farm.

The solution offered was to develop an autonomous beekeeping monitoring and control system and evaluate its operation in the real environment with the aim to increase the productivity of the beekeeping sector, reduce the unproductive work of beekeeping farms and improve the conditions of beekeeping.

Latvian scientists, technology companies and beekeepers have jointly developed an autonomous beekeeping solution that makes it possible to observe a beehive via a computer or smartphone [4; 7]. Unlike other solutions used in beekeeping in the world, this solution provides an opportunity to observe not only an individual hive, but the entire apiary.

By implementing an autonomous beekeeping solution in their apiaries, beekeepers will be able to remotely monitor critical hive conditions, such as temperature deviations in beehive, or if there is a lack of food in the flock, therefore, it will be possible to notice and prevent risks in time. Such system will help to collect honey in a timely manner, including monoculture honey, to track the living conditions of the flock and to detect bee diseases or parasites in the hive in time, as well as to save wintering bee colonies, save time and cost of transport in daily hive surveys. The developed solution will help to find out the connection between a better honey production place, flowering periods of various plants and nectar production; to determine the most active bee colony periods and weather conditions for effective honey production, to identify flowers flowering time necessary for monoculture honey collection.

The benefits of developed autonomous beekeeping systems are the remote monitoring of the production conditions in place, as well as timely identification of different negative impacts, e.g., ticks. According to Pudmeda [14], about 160 species of ticks live in bee colonies. According to Agrostory (2018), the tick of Varroa is a threat to honeybees. However, there are not only Varroa mites who make problems for beekeepers in apiary.

II. TECHNICAL SOLUTION

The Autonomous Beekeeping project took a step forward and developed a remote bee colony monitoring system that allows analysis at the apiary level and provides remote video monitoring in addition to mass, temperature, and meteorological data. The developed equipment has a vision for the future, based on IoT concept application [2; 10; 18].

The monitoring system developed in the project consists of five continuous weighing scales, indoor and outdoor temperature sensors, sensor wireless transmitters, four video cameras and a central control unit. The device allows remote retrieval of data both on the phone and with web applications. We also investigated the cybersecurity aspects [13] and autonomous power supply possibilities.

The monitoring system is in two apiary holdings: the first one in Riga, Latvian University Botanical Garden and the second one at Talsi region, Lubes County. In each apiary, we conducted monitoring in five beehives. We choose to use Norwegian multi-hull hives. Of course, it is technically possible to use any type of hive, but we preferred a multi-hull hive, as a hive with a perspective.

The beekeepers can use modern materials, methods, chemicals, instrumentation, and methodologies to make the process sustainable. Remote bee apiary monitoring with smart technology developed at the Autonomous Beekeeping project presented at Fig. 1, enable monitoring at various levels:

- At the individual insect / bee level.
- At the family / beehive level.
- At the population / apiary level.

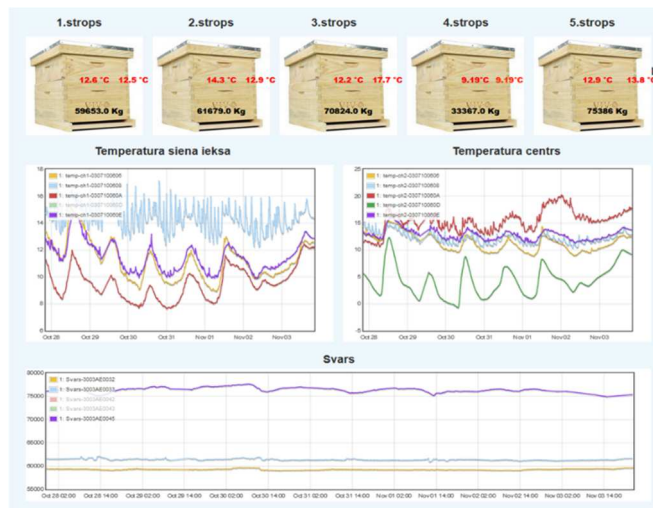


Fig. 1. Remote insect monitoring with smart technology is our „today“.

The monitoring system solution offers such opportunities:

- Temperature in two places inside of a beehive and one outside of a hive (temperature sensors).
- Hive mass.
- Relative humidity inside of a beehive.
- Video surveillance (at the entrance of each hive and one video camera for monitoring of a whole bee apiary).
- Meteorological data at a bee apiary (wind strength, direction, temperature, precipitation, RH%). We installed a meteorology station and aggregated its measurements with the data obtained by sensors of the monitoring system.
- Other.

Additionally, we installed a battery of accumulators, a wind generator and other equipment at the bee apiary in Botanical Garden to investigate conditions for autonomous

power supply of mobile apiaries, which should be reallocated 4-5 times during harvesting period. However, this part of the research is out of scope of this paper.

A. System installation

We implemented monitoring system installations in the following steps:

Step 1 - Levelling the base of the hive and positioning of scales (Fig. 2.)

Precision of instruments for weighing $\pm 10g$; weighing range up to 300kg; we have equipped 5 hives.



Fig. 2. Levelling the base of the hive.

Step 2 – We disassembled the Norwegian multi-level hive to its individual details. After that we inserted temperature and humidity sensors-transmitters, and their associated wires (Fig. 3.).



Fig. 3. We are inserting temperature sensors and their associated wires.

Step 3 - Positioning of temperature sensors.

Step 4 - Adjust the hive - return it to its original position.

Step 5 - Attach the sensor transmitters to the bottom housing or floor.

Step 6 - Prepare the mounting points for the camcorders.

Step 7 - Installation of outdoor camcorders at the entrance of the hives.

Step 8 - Installation of the outdoor camcorder - the apiary surveillance camera. Installation of recording equipment.

Step 9 - Locate the control unit / system in the apiary. Installation of a GSM router. We connected the control unit and the router to the electrical network.

Step 10 – Resolving a problem with cellular network coverage: amplifier antenna installation.

Step 11 - Data Retrieval to Mobile and WEB applications (Fig. 4).

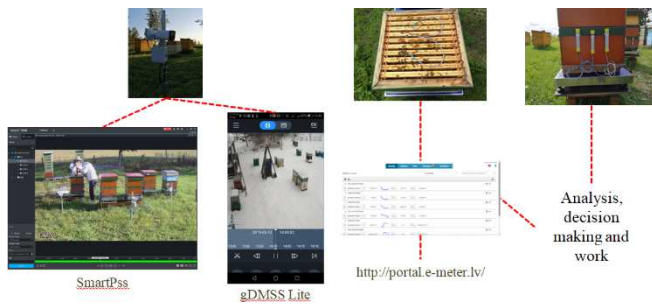


Fig. 4. Installation of the control unit.

III. USE OF VIDEO IN APIARY MONITORING

Another innovation is to analyze video files. The system has one common control [22] unit that concentrates information / data from five hives simultaneously (mass and temperature sensors) and four video cameras. [6] also have discussed the data security and reliability questions.

Monitoring five bee colonies in an apiary holding has significant advantages compared to monitoring one hive when we are monitoring a single flock or a group of bee colonies that can serve as a reference point. There is a possibility to compare between flocks, which is not possible in case of monitoring one bee colony.

Undoubtedly, the most interesting and innovative part of the system is the video equipment, which allows us to conduct video monitoring in the apiary. At the same time the video system has a heavy burden - the system must provide a large data flow.

We provided video surveillance by three hive close-up cameras and one apiary overview camera. We equipped the cameras with a remotely adjustable zoom, which is an especially useful feature for remote monitoring of bee colonies. Additional cameras provide night and day video surveillance modes. For remote video surveillance of beekeepers' apiaries and beehives online we can use both phone and computer.

Compared to the information provided by mass and temperature sensors, video monitoring provides the beekeeper with much more extensive and detailed information.

We present existing possibilities to use of video monitoring to make the apiary sustainable; more significant for practical beekeeping are:

1. Diagnosis of diseases.
2. Effects of pesticides on the bee colony.
3. Animal damage diagnostics.
4. Pollen / propolis collection.
5. Flight intensity.
6. Metrological data.
7. Robbery.
8. Compassion.
9. Protection of the apiary.

Video monitoring gives us an advantage in diagnostic diseases. Currently, no monitoring system available on the market offers remote diagnostics of bee colony diseases. The monitoring system developed by autonomous beekeeping proves in practice that video monitoring at bee colonies can diagnose following diseases, such as nematodes, deformed

wing virus (DSV), Varroa mite, lime stumps and other peruvian diseases that a beekeeper can clear from bees. Video monitoring (with additional algorithms of object recognition) can be a great tool, which gives the beekeeper the opportunity to react to the threat as fast as possible and in an adequate manner.

IV. QUALITY-DRIVEN DECISION-MAKING TECHNIQUES TO IMPROVE RESILIENCE OF WIRELESS COMMUNICATIONS

We evaluated the first prototype against transition reliability parameters. During the testing periods there were situations when heavy rain, dense fog, snow, frost, and extreme temperatures took place, which has a considerable influence on reliable communication over networks. A wireless link is vulnerable to a variety of precipitations, a level of humidity as well as to fluctuations caused by reflections of signals from moving objects. This means, that wireless signal can experience so-called "path loss" or attenuation of signal strength [11; 12]. In this case, critical environmental changes in communication and its degradation are noticeable by users as well as network operators while only network operators perform evaluation of service and network quality. We have analyzed the quality of received signal using detected signal attenuation, and signal losses: Quality of Delivery, Quality of Experience and their correlation with the Quality of Protection and Quality of Resilience of the prototype network. The degradation of parameters describing an infrastructure, i.e., a transmission system, and devices, which participate in a service transmission chain have an influence on QoS. The factors which influence the Quality of Delivery (QoD) are related to the communication process and data parameters. This means that all the characteristics in the communication process, like routing protocols, path, data processing and recovery mechanisms impact QoD [15]. We did the perceptual evaluation of the QoS, service completeness, cost/service ratio during testing of the prototype of the monitoring systems.

V. TESTING RESULTS

Hygienic behavior characterizes the ability of the bee colony to keep the nest clean and to recognize dead flocks and clean them from the nest. In practice, we are checking the hygienic behavior of the bee colony either by a needle or by freezing.

To analyze the application of video monitoring in the diagnosis of root diseases and hygiene behavior, we conducted a needle test in separate bee colonies within the project. The goal is simple, to diagnose fragments of dead Peruvian bodies at the run. The study culminated in success, in the bee colony performed in the needle test, close-up video cameras show nest common bees, which take out small pieces of dead brood bodies in the upper jaw. Later in the autumn, we observed remarkably similar behavior in bee colonies, where we diagnosed a high tick population and the presence of DSV (note, these bee colonies have dead brood).

The use of agrochemicals has a wide range of negative effects on beekeeping. The problem culminates when a beekeeper observes the death of bee colonies or sees dead flying bees on the run. More often in practice we have the second scenario - flying bees die at the ranks of bee colonies. In a situation where flying bees come into contact with an active substance that is toxic to bees, there are two possible scenarios - (i) the flying bee dies in the field and does not return to the nest and / or (ii) returns to the hive and ends either on the run or on the floor. It is possible to visually diagnose dead bees at the beehive, especially if the hive has a wide treadmill, a pollen receiver or a larger plate placed on the

ground. Dead bees in the hall are hard to spot! As a result of the direct negative effects of pesticides, we can observe dozens or even hundreds of working bees at the bee colony. They are characterized by a complex set of features - (i) these bees do not fly; (ii) part of the bees are dead; (iii) individual bees chaotically crawl; (iv) others are lying on their sides and impulsively twitching, and finally a small proportion of flying bees lie on their backs with their feet upside down and twitching limbs (see Fig. 5). Pesticides have damaged bees' nervous system. The beekeeper can observe such signs in the apiary for only 2-3 hours because the bee colony rapidly loses flying bees, and the nest loses the transmission of information about the location of the food base (the dance ritual about the location of the dangerous food base is over).



Fig. 5. Pesticides are affecting flying bees. Video taken in Denmark in July 2019.

Often, the beekeeper does not observe this situation and only during the next bee colony inspection, does he find that the flock is falling in force and size, thus failing to explain the cause. Remote monitoring of bee colonies can significantly improve the situation, allowing remote diagnosis of this type of problem in the apiary.

Different animals can endanger bees' colonies and apiaries. We can easily diagnose their presence or damage by video surveillance. Let us look at the following examples.

(i) Bears. In recent years, the amount of bear damage in apiary holdings has increased significantly in the eastern part of Latvia. This summer, bears have destroyed more than ten apiary holdings in Latvia.

Ari Seppälä, a Finnish beekeeper and expert on bears, says: "We divide bears into two groups. One group consists of inexperienced animals and the other of experienced bears. Inexperienced bears visit the apiary 3-4 times without touching the bee colonies, sometimes expelling a flock, but do not damage the apiary. However, peace does not last forever, soon the inexperienced bear learns a trade and clings to bee colonies. The second group consists of experienced bears. Experienced bears visit the apiary every day and destroy bee colonies one by one. All bears behave and their technique is similar. They overturn the hulls of the hive one by one and, when rolled on the ground, they dump the hoppers. Next, the bear is lifting dumped hoppers one by one and goes 50-100m from the apiary, where it condemns the gain. The bear continues until destroy the entire apiary."

Often the beekeeper finds out what is happening in the apiary holding late, when the apiary has already suffered extensive damage. Remote video monitoring allows the beekeeper to identify problems in a timely manner and take the necessary measures to protect the apiary, such as relocating the apiary.

(ii) Birds. Certain bird species that feed on live or dead bee individuals, such as pupils, are a threat to wintering of bees' colonies.

If the bird population in the vicinity of the apiary is large, the damage can be significant. In addition, with their activity

and knocking, birds cause stress to bee colonies, which negatively affects the quality of wintering.

For beekeepers working with foam hulls, woodcuts and woodpeckers also cause physical damage to the hive. The apiary overview video allows you to freely diagnose the extent of the bird problem and eliminate it in a timely manner.

(iii) Marten. The agile and small animal tends to damage both the hives and the bee colonies. We have detected the problem with a marten during video surveillance.

(iv) Wasps. They are dangerous to bees' colonies in late summer and autumn, especially for problematic and weaker bee colonies. Sometimes wasps can destroy the bee colony.

Close-up video surveillance of the hive run allows to diagnose the activity of wasps near the bee colony. If necessary, the beekeeper can narrow the run.

Among the wasps are also the European hornet, which is a pronounced predator of flying bees. European hornet activity in apiary holdings increases in the autumn. In case of a problem, video monitoring allows us to diagnose it.

A. Analysis of acoustic signals

The future steps of prototype development are related with acoustic data collection, which allows us to deduce the rhythm of hive life remotely, using sound, which allows us to identify in a timely manner and eliminate existing problems.

The singing of the queen bee is the acoustic noise made by the queen before leaving the second colony; the behavior of the mother is determined by its frequency of sound. Most bee colonies swarm from late May to early July. The pressing time lasts no more than 30-40 days. However, the squeezing time of bee colonies depends mainly on the time of the start of the main honey harvest, the natural and climatic conditions of the area, the condition of the flocks after wintering and the bee breed [19; 20; 21].

The bee colonies are in a pruning position 7-10 days before the first flock. If you do not respond appropriately to prevent further swarming, this family condition can last for another 10-15 days. It stops after the bees destroy all the mating mothers, leaving only one new mother [8].

Active modernization of beekeeping now is underway, the aim of which is to facilitate the work of beekeepers in servicing 50-100 of beehives. The beekeeper will receive information about the acoustic noise of each hive and visualize it all on a single graph or computer table, providing a simple data analysis system.

Nowadays, acoustic signal analysis and processing often provide in the following sequence [26]. We acquire the sound inside or outside the hive using microphones. Then, we filter or resample the sound signal to remove noise and spare frequencies. After that, we extract necessary features from the signal, based on different algorithms, e.g., Principal Component Analysis. Then, we use the classification algorithms to detect the colony health status, such as Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and Machine Learning (ML).

Cejrowski et al. [5] proposed a method to detect the presence of the queen bee based on sound analysis. The system includes a custom brood frame placed inside the beehive, with a microphone, and a temperature and humidity sensor. They created a critical situation by removing the queen bee from one of the hives. They analyzed data acquired with the normal colony and the orphaned colony using Linear Predictive

Coding (LPC) for features extraction. Kulyukin et al. [9] published an article where they exploit ML techniques to analyze the sound of the hives. The main goal of this research was to distinguish the honeybee sound, from the background noise and the cricket chirping noise. The researchers used microphones placed outside the entrance of bee hives.

The paper [17] demonstrates long term study automated on in-situ non-invasive monitoring of a honeybee vibrational pulse signals using ultra-sensitive accelerometers embedded in the honeycomb located at the center of beehive. Unlike microphones, the sensors-accelerometers exploit vibrations and do not suffer from propolization problems. From the sustainability point of view, application of long-term operating sensors looks more promising for using them in an autonomous beekeeping system. That is why we will focus our further research on acoustic signal analysis on such type sensors.

One of the key features of a diagnostic system is its reliability [16; 22]. We chose a microcontroller as the hardware and technical base of the system, which contains modern protection against failures. An important criterion of the diagnostic system is the diagnosis of malfunctions and the capabilities of the monitoring system. During operation, unforeseen situations may happen, the cause of which may be climatic conditions, or the cause of which may be external mechanical influences. In the event of an error from any element, the system executes an alarm message.

VI. CONCLUSIONS

System solutions use not only video camcorders, but also different sensors to collect all the necessary information for sustainable autonomous beekeeping system.

In this article, we have looked at examples of video applications in bee colony monitoring. Combination of video, temperature, humidity, mass measurements and other data offered by the monitoring system raises Autonomous Beekeeping to a new dimension. This system provides a wide range of possibilities for remote monitoring in practical beekeeping as well as in science.

Video monitoring is very promising - the beekeeper provides clear and easy-to-interpret information. At the same time, video is also a shadow side – the beekeeper must analyze online or archive files manually, which takes time. Engineers are optimistic. Video analytics and processing tools are already available, and they allow detecting certain events automatically. Beekeepers must keep up with how video surveillance develops in the future.

Autonomous beekeeping allows beekeepers to save resources and increase the amount of their product. The system for traditional beekeeping is something new. Our team member has elaborated and evaluated the master's thesis within the project "Autonomous beekeeping," we have improved the existing prototype.

We have developed a 3D-sun tracker model and evaluated and fully developed a prototype. We have developed the operation algorithm of the tracking device; we have written the program "Atmega 328" for the microcontroller. The prototype given is ridiculously cheap, in total it cost forty euros. We concluded that the use of a solar tracker allows electricity production to increase by 15%, regardless of the outdoor conditions. For example, we have obtained from existing results that an average we generated of around 900 Wh in just one day. Installing solar trackers with solar tracking we have generated around 1035Wh. Setting the solar tracker

to solar tracking one time to 1 hour to save electricity. The total loss for a solar tracker per day is around 10-25 Wh depending on the servomotors.

In this paper we investigated literature devoted to the use of acoustic sensors and sensors-accelerometers. We made a conclusion that from the sustainability point of view, application of long-term operating sensors looks more promising for using them in autonomous beekeeping system. Therefore, we will focus our further research on acoustic signal analysis on such type sensors. The introduction of sensor-accelerometers in the monitoring system will allow us to expand the information about the condition of the apiary.

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